SOLUTION DIAGRAM & ARCHITECTURE

Turn retiring experts’ judgement into rules AI can’t override.

Keystone compiles what a senior engineer knows into an **Intelligence Pill** — a versioned, tested, plain-text artefact the organisation owns, with deterministic gates no model can talk past.

`keystone-tencent.vercel.app`
live demo · passcode `keystone`

`github.com/frederickemerson/keystone`
source, pills and tests

THE REAL-WORLD PROBLEM

The judgement that keeps a building safe is in a few heads, and those heads are leaving.

1

Concentrated

Keppel's own brief: critical knowledge sits with a few experts; new staff need a long time to build it.

- One Chief Engineer, 18 years of M&E, holds the chilled-water leak playbook.
- A wrong call means unsafe re-energisation.

2

Never written down

The useful part is tacit: when to break the rule, and why.

- "Always isolate the riser" **vs** "isolate the sub-valve" — both true, under different conditions.
- SOPs record the step, not the judgement.

3

Lost at retirement

Replacing an employee costs 0.5–2× annual salary (Gallup); FM talent growth "is not keeping pace" (JTC, 2025).

- The knowledge leaves with the person.

4

AI can't be trusted to act

A chatbot that answers from a manual will also answer when it shouldn't.

- Asked outside its pill, our published agent replies: *"This is not covered by an approved pill."*

Who this is for

REIT managers and property/facility-management organisations running mixed portfolios — offices, data centres, logistics. We model the real chain: REIT manager → property/facility manager → FM contractor, with 15 named personas.

Pain points sourced from the Keppel brief, Investor Day 2026, Keppel REIT SR24, Gallup and JTC

Keystone

Spoken judgement becomes a governed Intelligence Pill

Solution diagram · Keppel AI HARVEST
Pill in focus: Technical Services, chilled-water leak · fictional data

1 Business problem

Incident-response judgement sits with a few senior engineers, is applied inconsistently and leaves when they retire. A wrong call on a leak means unsafe re-energisation, tenant/SLA exposure and downtime. A chatbot can't be trusted to act.

2 Capture what was never written

Whose: Tan Wei Ming, retiring Chief Engineer, 18 years of M&E.

- Onboarding call with **CDM + ACTA probes** (simulated in the demo): "walk me through the last time…"
- **Contradiction resolution:** "always isolate the riser" vs "isolate the sub-valve" becomes one condition (TS-02)
- **Expert confirms, edits or rejects** every rule on the Decision Canvas

3 Compile into a reusable pill

A plain-text package the organisation owns:

`SKILL.md` `rules.yaml` `evidence.yaml` `provenance.json` `evals/` `CHANGELOG.md`

- 13 rules, each with action class, owner, deadline and expert quote + timestamp
- Owner, steward and approver named per pill
- Runs as a **native CodeBuddy skill** (verified): same TS-07 decision

4 Asset Operations Manager supervises

Incident Room: a deterministic engine applies the pill to evidence.

Recommend · Requires approval · Must escalate · Blocked

- **Verify:** Why drawer shows rule, version, missing evidence and the expert's quote, read aloud
- **Escalate:** TS-07 blocks re-energisation and routes to the duty LEW by name
- **Approve:** named humans only; server-side role-based redaction

5 More than a chatbot

- **Deterministic gates** an LLM cannot talk past
- **Provenance** to the spoken sentence
- **Evals** replay past incidents
- **Governance:** versions, owners, audit, rollback
- Engine takes no model input: an LLM change can't alter a gate

6 Governed learning after deployment

Evals pass + an approver who neither wrote nor shadowed it signs. History is append-only.

Feedback	Review	Diff	Validate	Shadow	Approve	Release
Outcome captured: isolated in 7 min	Expert corrects TS-02 deadline 15 → 10 min	v1.2.0 → v1.3.0 draft, reason recorded	Replay evals on past incidents; regressions block	Steward replays 6 closed incidents on both versions: 6/6 agree	Three people: author ≠ steward ≠ approver; enforced in code	New semver version; one-click rollback

7 Extend across assets and units

- **Asset overrides:** on Harbourfront DC1 the same pill escalates to the Critical Facilities Manager
- **Coordinate:** Tenant Experience pill runs beside Technical Services
- **Registry:** coverage and gaps by asset type and domain

8 Credible implementation path on Tencent Cloud

Running today: Next.js app on Vercel (live AI there: hy4-preview via TokenHub, then OpenAI, else recorded), deterministic engine, CodeBuddy skill (verified)

CodeBuddy / WorkBuddy	Hunyuan models	ADP	iBMS · CMMS	TRTC voice	EdgeOne · SSO
Pill as SKILL.md skill (verified); WorkBuddy Experts proposed	hy4-preview via TokenHub (live, ap-singapore) and CodeBuddy SDK (local)	Proposed: governed skill distribution, Approval Center	Proposed: leak, alarm and work-order feeds (Brick 1.5); mock today	Proposed: live onboarding call with captions	Proposed hosting (Vercel today); enterprise SSO for roles

BUSINESS ARCHITECTURE

One loop, five stages, a named human at every gate.

Capture

Teach a skill

The expert answers one wide question, then only what is still missing.

- Answers are taps, so a **contradiction is arithmetic**, not a model's opinion.
- A frontline note at the moment a gate blocks someone feeds the same loop.

Tan Wei Ming · Rajesh Kumar

Compile

Confirm the rule

The AI only drafts. Every rule shows the sentence it came from, with a timestamp.

- The expert confirms, edits or rejects each one on the Decision Canvas.

Tan Wei Ming

Apply

The gate holds

A deterministic engine applies the pill to the evidence of a live incident.

- TS-07 blocks re-energisation and escalates to the **duty Licensed Electrical Worker**, by name.

Priya Nair

Learn

Correct it safely

Diff → eval replay on past incidents → shadow → approval → one-click rollback.

- **Three different people**: author ≠ steward ≠ approver, enforced in code.

Hafiz · Priya · Goh

Reuse

Across the portfolio

The same pill runs at a colocation data centre through asset overrides.

- Escalation reroutes to the Critical Facilities Manager; the notice becomes a client-SLA notice.

Every asset of that type

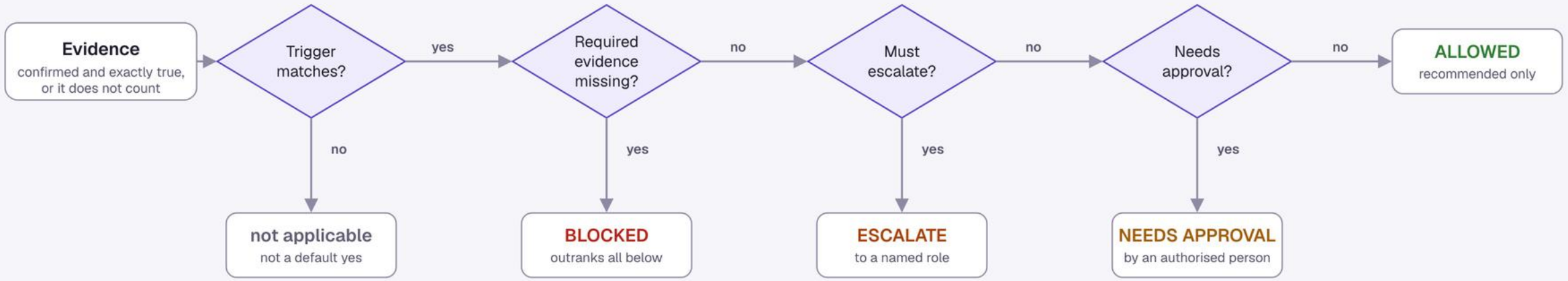
The loop closes

Reuse feeds the registry, which shows coverage and gaps by asset type and domain — so the next gap is visible before the next expert retires. Every stage in this row is built and working in the demo; nothing here is a mockup.

Guided demo walks all five stages, one person and one screen per step

FLOW 1 · THE DECISION PATH

How a gate decides — and why a model cannot move it.



Asset overrides are applied **before** the status is computed — the same rule escalates to the Critical Facilities Manager at a data centre.

An exception may **add** required evidence or change who it escalates to. There is no path by which an exception relaxes a gate.

The evidence

Water reached DB-23A and the circuit is isolated — but no Licensed Electrical Worker has inspected it yet.

The rule fires

TS-07 triggers. `lew_signoff` is required, absent, and the rule forbids the action without it.

The outcome

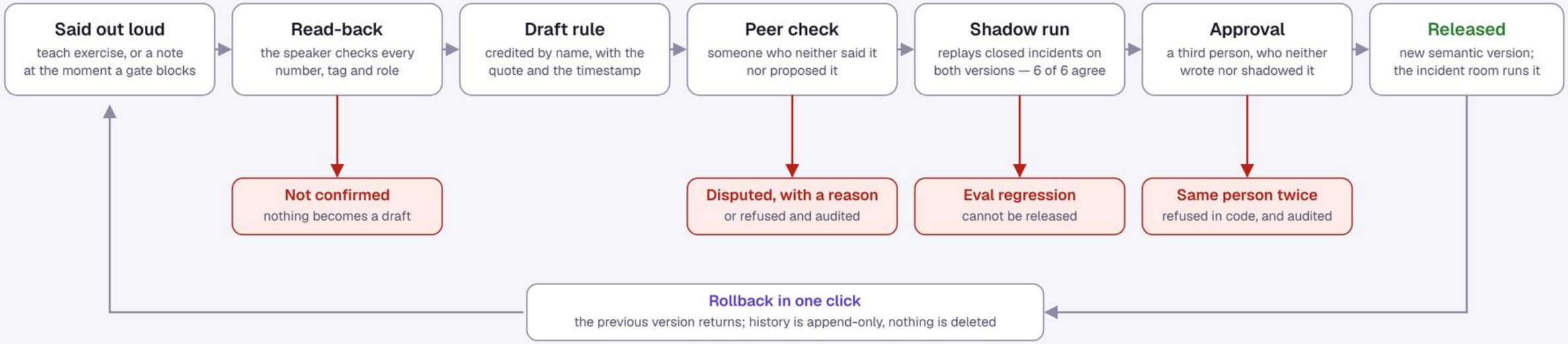
Blocked. Escalated to the duty LEW by name. The AI's proposal to re-energise AHU-23-01 is held, not executed.

Whatever the outcome, nothing executes

Every branch of this chart ends with a person. The engine is pure TypeScript with no model input, no `eval()` and no string expressions — so changing the model, or the prompt, cannot change which branch you land on.

FLOW 2 · GOVERNED LEARNING

From one spoken sentence to a released version — and back.



Author ≠ shadow promoter ≠ approver. The code refuses the second role to the same person and writes the refusal to the audit log.

What the demo actually walks

Rajesh's twenty-second note at the blocked gate becomes DRAFT-01, credited to him. Hafiz confirms it — he neither said it nor proposed it. Priya puts it into Shadow and replays six closed incidents; six of six agree. Goh approves, and the incident room runs the new version, one click away from rollback. Four people, four screens, every refusal audited.

This is the part that is usually a slide

Separation of duties, eval gating, shadow mode, audit and rollback are built and enforced here — not described as a future control. The guided demo walks a real correction through every box on this chart.

THE REUSABLE UNIT

An Intelligence Pill is plain text the organisation owns.

What is in the package

`pills/technical-services/chilled-water-leak/`

`SKILL.md`	Owner, steward, approver, version, access and boundaries — plus the prose an LLM host reads.
`rules.yaml`	**13 rules**: structured triggers, required evidence, action class, forbidden actions, escalation target, deadline, asset overrides.
`evidence.yaml`	The shared vocabulary — one contract both the pill and the app are tested against.
`provenance.json`	Every rule mapped to the expert's quote, timestamp and confirmation.
`evals/`	Six past incidents a new version must still agree with.
`CHANGELOG.md`	Append-only history.

Why plain text matters

- **Portable.** Copied unchanged into `.codebuddy/skills/`, CodeBuddy reached the same TS-07 decision as our engine — blocked, escalate to the duty LEW, quoting the Chief Engineer. **Verified, not claimed.**
- **Reviewable.** A rule change is a diff an engineer can read.
- **Model-independent.** The decision path takes no model input, so swapping the model cannot change a safety outcome.
- **Yours.** It is a folder in your repository, not rows in a vendor's database.

The gate, in one line

Status precedence is fixed code:

not applicable → blocked → escalate → needs approval → allowed

- Evidence counts only when it is confirmed and exactly true.
- An exception may **add** a requirement. It can never relax a gate.
- No AI proposal ever executes. Each is held for a named human.

Two pills, one incident

A second pill — `tenant-experience/incident-comms` — runs beside it during the same leak: it drafts the tenant notice under its own hard rules while Technical Services works the gate.

TECHNICAL ARCHITECTURE

Two paths, one line between them.

Surfaces · Next.js 15 App Router, React 19, Tailwind v4

Teach a skill · Expert call (recorded) · Decision Canvas · Incident Room · Pill Registry · Approvals · Outcome & Learning · QR capture at the plant

Server · route handlers and middleware

Default-deny middleware with a public allowlist · same-origin checks on auth · CSP and security headers · per-session and per-IP rate limits · role read from a signed session, and data above that clearance masked before it leaves the server

no model output crosses this line

Decides · deterministic

lib/engine

Pure TypeScript. No LLM, no eval(), no string expressions.

Conditions: eq, neq, gt, gte, lt, lte, in, exists.

Returns actions, an evidence count and a trace — plus counterfactuals and evals.

lib/governance

Semantic versions, JSON diff, append-only rollback.

Three-person separation of duties, enforced; refusals written to the audit log.

Eval-regression gate before any release.

pills/ + data/

The content, as plain files.

One shared contract per pill — evidence.yaml — and a test that fails on any key outside it.

Versions, approvals and the audit log are browser-local in the demo; server-side in production.

Writes language · the model

lib/llm

Hunyuan hy4-preview via Tencent TokenHub — dedicated Singapore endpoint, ~3.9 s, thinking off, strict structured output.

Then the CodeBuddy Agent SDK (locally), then OpenAI, then recorded output — and the badge always names who answered.

lib/speech · lib/capture

MiniMax Speech 2.8 HD via TokenHub reads the expert and the interviewer aloud; lines are pre-rendered so a demo never waits on a quota.

Capture: consent, hold-to-talk, read-back chips found by regex and the registers — never the LLM.

Running today, and what comes next

Live on the deployed site: the app on Vercel, the deterministic engine, the pill as a native CodeBuddy skill, and the MiniMax voice on screen. Integrated through Tencent TokenHub on a Singapore endpoint: Hunyuan hy4-preview for extraction, MiniMax Speech 2.8 HD for synthesis. Proposed: Tencent Cloud ASR in Singapore, ADP skill distribution, TRTC for the live voice interview, iBMS and CMMS as evidence feeds, EdgeOne Pages for hosting, enterprise SSO for roles.

Deployed on Vercel today; Tencent EdgeOne Pages or Lighthouse is the proposed host

HOW PROMPTS DRIVE THE AI GENERATION

The model does language. It never decides a safety action.

1

Elicit

One wide question first — "walk me through the last time…" — because an unprompted account gives up more than ten narrow questions do.

- Probes tagged against a 31-probe library from the Critical Decision Method and ACTA.

2

Extract

A schema-constrained "scribe" prompt maps speech into the `rules.yaml` schema.

- Strict structured output, thinking disabled.
- Output is always a **DRAFT candidate** the expert must confirm.
- If no live model answers, it says so — it will not show a recorded reading of a different conversation.

3

Resolve

Two answers about the same field with different claims disagree.

- In the exercise this is **string comparison, not inference** — the engine simply knows.
- The conflict becomes one distinguishing condition; neither half is dropped.

4

Compose

Tenant notices are drafted under the Tenant Experience pill's hard rules.

- No cause before the cause is confirmed (TE-03); no completion time before isolation (TE-02); PII masked (TE-05).
- Unsupported prose fails closed. A named human approves. Keystone never sends it.

The invariant

The decision path takes no model input. Change the model, change the wording — you cannot change whether the gate holds.

Rule drafting, contradiction resolution and notice drafting are all demonstrable in the live app

BUILT AND WORKING

Features we would show you, not describe to you.

Teach any pill

A situation, one open question, then only what is still missing — and a board showing which fields of the rule are still unknown.

- Questions for a pill nobody wrote any for are **derived from that pill's own rules**, wrong answers included.

Hear where it came from

Every rule links to the second of the recorded call where it was said, and plays it.

- Provenance is a quote, a timestamp and the name of whoever confirmed it.

Why? and what-if

Each decision opens a drawer: rule, pill version, evidence used, evidence missing, owner, escalation target.

- **Counterfactual:** "what would change this decision?" is answered from the unmet conditions, not guessed.

Capture at the plant

A QR label on the equipment opens a capture task. Consent first; audio is not kept; credit is opt-in.

- Read-back chips for numbers, times, tags and roles — found by regex and the registers, never by the LLM.

Replay before release

A new version is replayed against closed incidents and reported as "x of y assertions" — a count, never an invented percentage.

One pill, many assets

Asset overrides reroute the same rule: at a colocation data centre TS-07 escalates to the Critical Facilities Manager and the notice becomes a client-SLA notice.

Roles that actually bite

Fifteen personas. The page reads your role from a signed session and the server masks anything above your clearance before it reaches the browser.

Runs inside CodeBuddy

The pill installs unchanged as a native CodeBuddy skill and reaches the same TS-07 decision as our engine. Verified, with screenshots.

And what we deliberately did not build

The teach exercise previews a draft and stops there — choosing which existing rule a new statement amends is a judgement, so a person places it. Hand-written questions exist for one pill; the rest are generated. Versions and the audit log are browser-local in this demo. Speech recognition has not been tested on real Singapore-accented field speech, and we do not claim it.

Guided demo at /guide walks the whole loop · every screen is live on the deployed site

WHAT IS NOVEL

Five things here that a retrieval chatbot cannot do.

1

A gate, not a prompt

Safety lives in fixed code, not in instructions a model is asked to follow.

- Exceptions can only **add** requirements — there is no path that relaxes a gate.
- Prompt injection cannot reach the decision: it takes no model input at all.

2

Contradiction as arithmetic

Each answer carries a claim token about one field of a rule. Two different claims about one field disagree — by string comparison.

- Nothing has to *notice* the conflict, so nothing can fail to.
- Both statements are kept; the condition that separates them is named.

3

Knowledge that outlives the model

The pill is plain text with an owner, a version and tests — not weights, not an index.

- The same file reached the same decision in CodeBuddy as in our engine.
- Switch vendor and the judgement comes with you.

4

It refuses to invent

Asked outside its pill, the published agent says *"this is not covered by an approved pill"* rather than answering.

- When no live model answers an extraction, it says so — it will not pass a recorded reading of another conversation off as a reading of yours.

5

Governance that executes

Separation of duties, eval gating, shadow mode, audit and rollback are code paths with tests, not a policy page.

- The author cannot shadow it; the shadow promoter cannot approve it; refusals are audited.

The test we hold ourselves to

Every claim on these slides is either demonstrable in the live app or stated as proposed. Where something is simulated — the onboarding interview, the incident data — the screen says so in words, on the screen.

1,302 automated tests across 103 files, including property-based tests on the engine and governance

BUSINESS VALUE

What it is worth, and how we would measure it.

0.5–2×

annual salary to replace an employee (Gallup) — and with them, the judgement nobody wrote down.

>US$100k

the cost of more than half of significant data-centre outages; about 1 in 5 cost more than US$1M (Uptime Institute).

S$125k

a month: one 10,000 sq ft Grade A floor at S$12.50 psf (CBRE Q2 2026). A two-week closure is ~S$58k in rent alone.

What Keystone already measures, over the demo data

- **Time to isolate**, read from the incident records.
- **Replay agreement** — "x of y assertions", from the deterministic engine, never a made-up percentage.
- **Eval pass and shadow agreement** per pill; decisions awaiting approval; pills approved; pending corrections.
- An illustrative **downtime-avoided** estimate, with its assumption stated on screen (4 hours per blocked re-energisation).

The pilot would measure

- Time to isolate · escalation latency · gate-block rate · **override rate** · eval agreement · rollback frequency.
- Goal: a measurable cut in time to isolate and in unplanned re-energisation risk — and new engineers making guided decisions with named escalation **from day one**, instead of after months of shadowing.

Anchored to Keppel's own goal

Technical Services aims to "cut downtime and costs" (Investor Day 2026). Keppel DC REIT reports ~12% better effective PUE on asset-enhanced colocation assets (AR2025) — the gains our data-centre overlay is built to protect.

Figures are external and cited; everything measured over demo data is computed from the records, not asserted